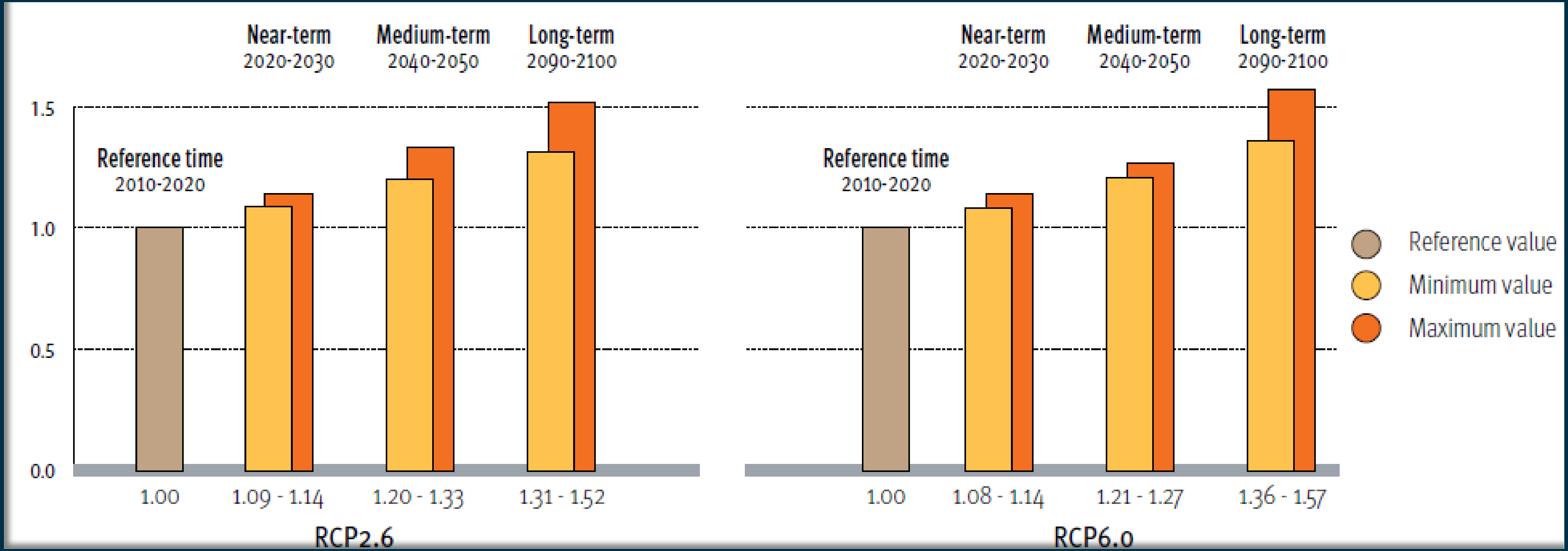
Uncontrollable and devastating wildfires are becoming an expected part of our seasonal calendars.

Wildfire is "an unusual or extraordinary free-burning vegetation fire which may be started maliciously, accidently, or through natural means, that negatively influences social, economic, or environmental values" Sullivan, Andrew, et al. (2022).

Climate change and land-use change are projected to make wildfires more frequent & intense.

By **2030**, a rise of up to **14%** in extreme fires globally is expected, and by the end of **2050**, this increase could reach **30%**.

Sullivan, Andrew, et al. (2022).



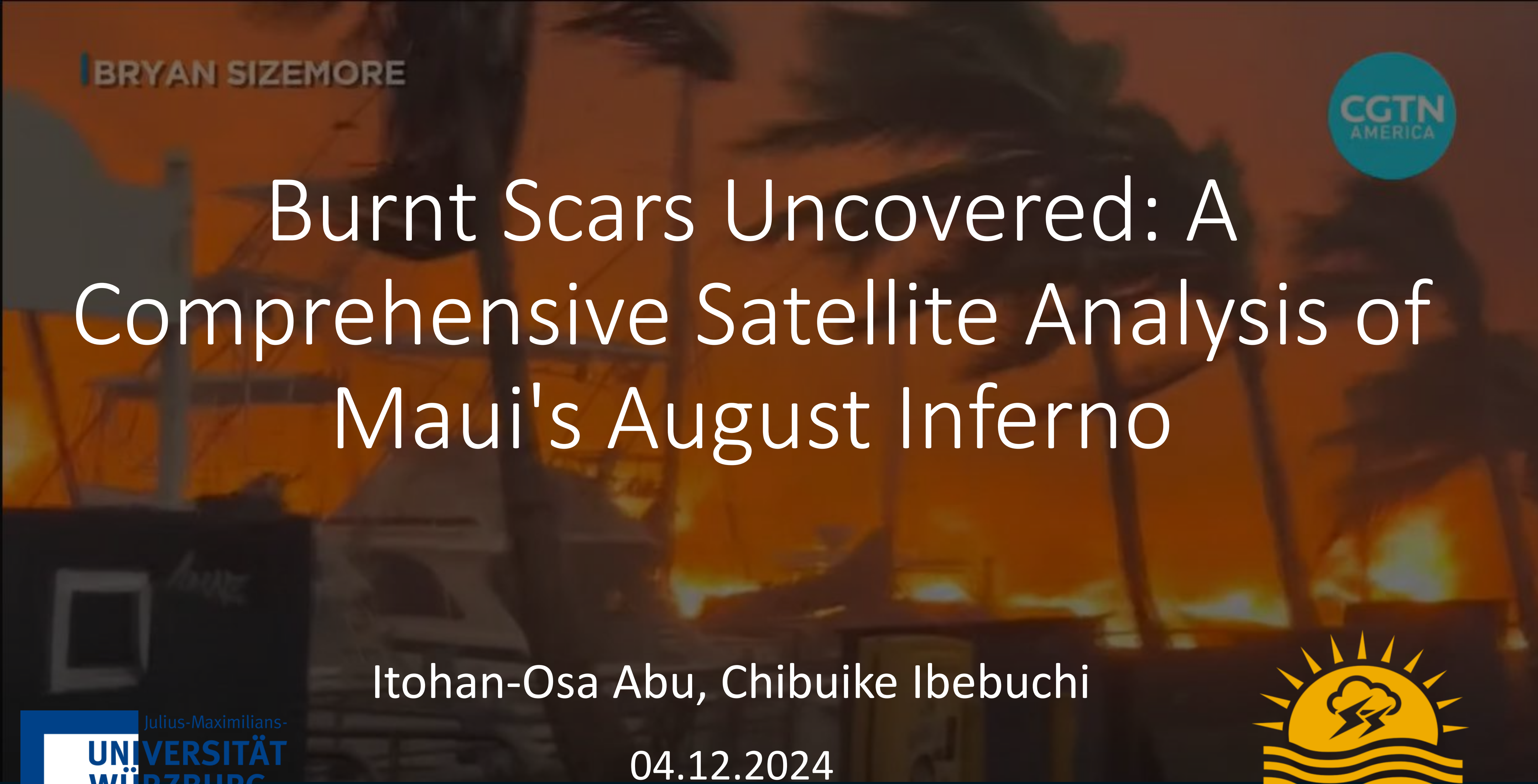
By the end of the century, the likelihood of catastrophic wildfires events will increase by a factor of 1.31 to 1.57.

Sullivan, Andrew, et al. (2022).

The differences between Wildfires & Landscape fires

Landscape fires	Wildfires
Often seasonal	Linked to extreme weather
Low to moderate intensity with short episodes of high intensity	Mostly high intensity with some periods of moderate intensity

Easily controlled with regular firefighting resources	Control measures may exceed regular firefighting resources
Low cost impact	High cost impact
Landscape fires	Wildfires



Burnt Scars Uncovered: A Comprehensive Satellite Analysis of Maui's August Inferno

Itohan-Osa Abu, Chibuike Ibebuchi

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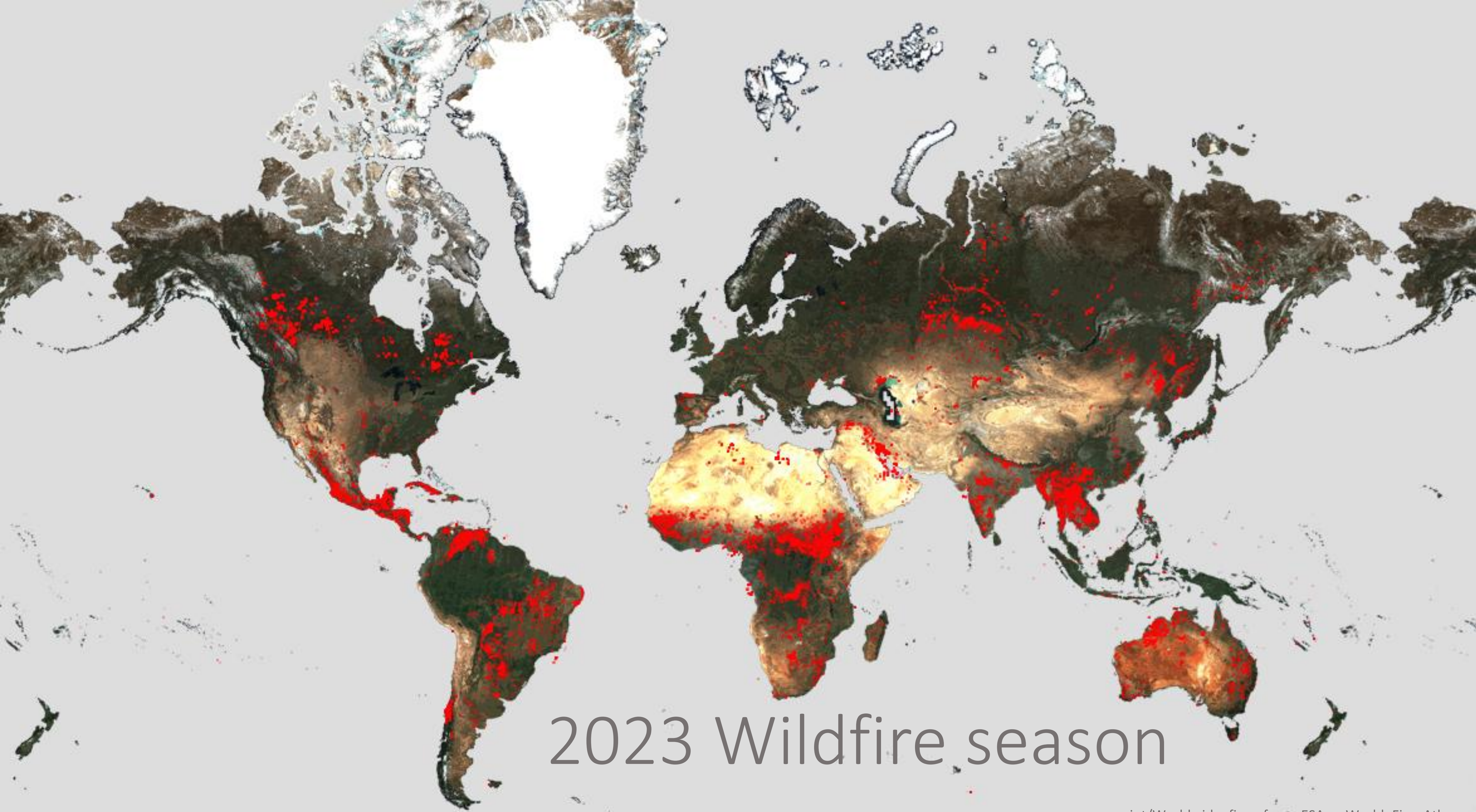
WÜRZBURG

CLIMRISE LAB
KENT STATE UNIVERSITY

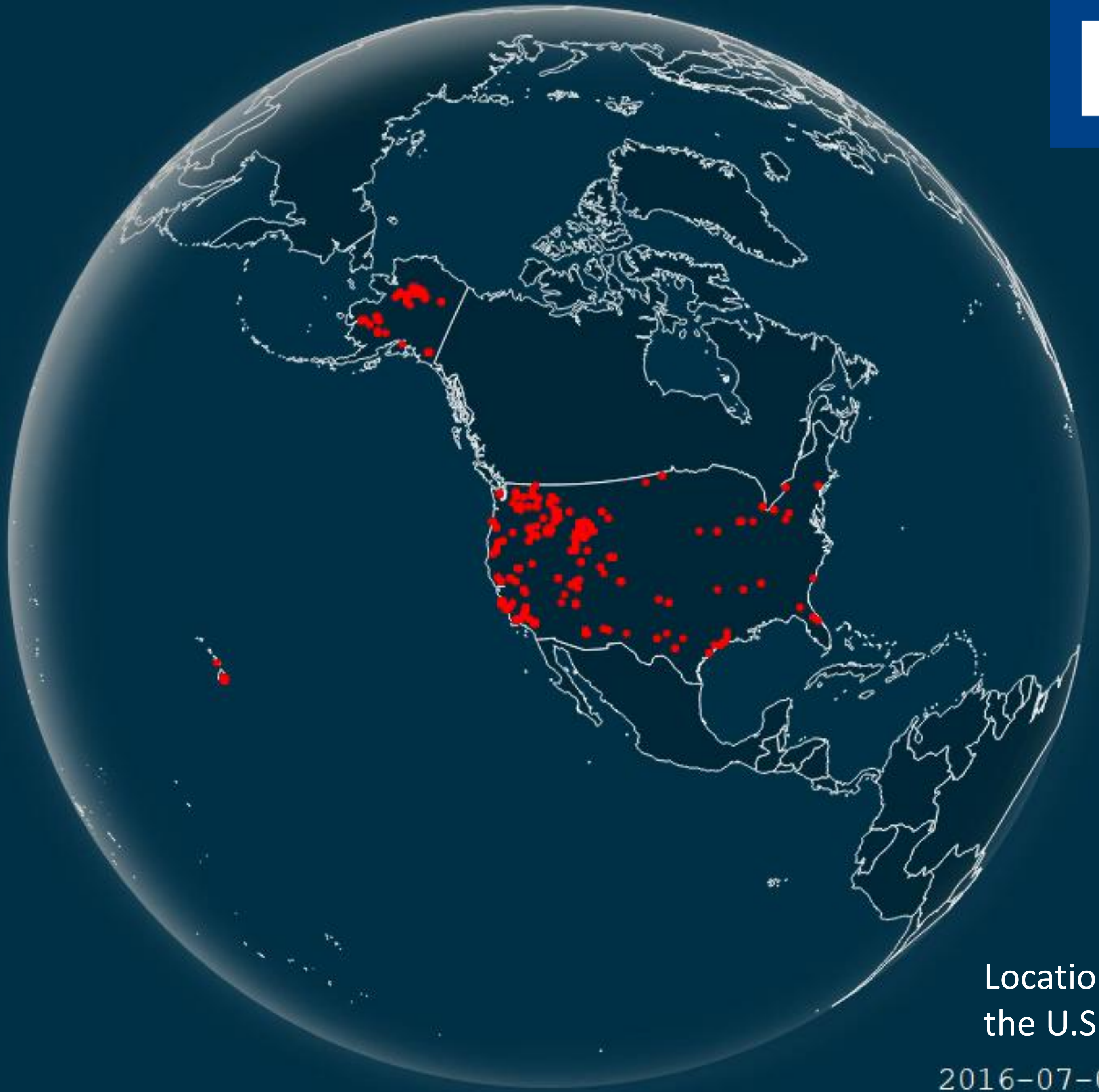
04.12.2024

2023 Wildfire season



www.esa.int/Worldwide_fires_from_ESA_s_World_Fire_Atlas

The U.S. has already seen a devastating string of catastrophic wildfires in 2023...



Locations of fire events in the U.S since 2016 till date

https://s3wfa.esa.int/dashboard.

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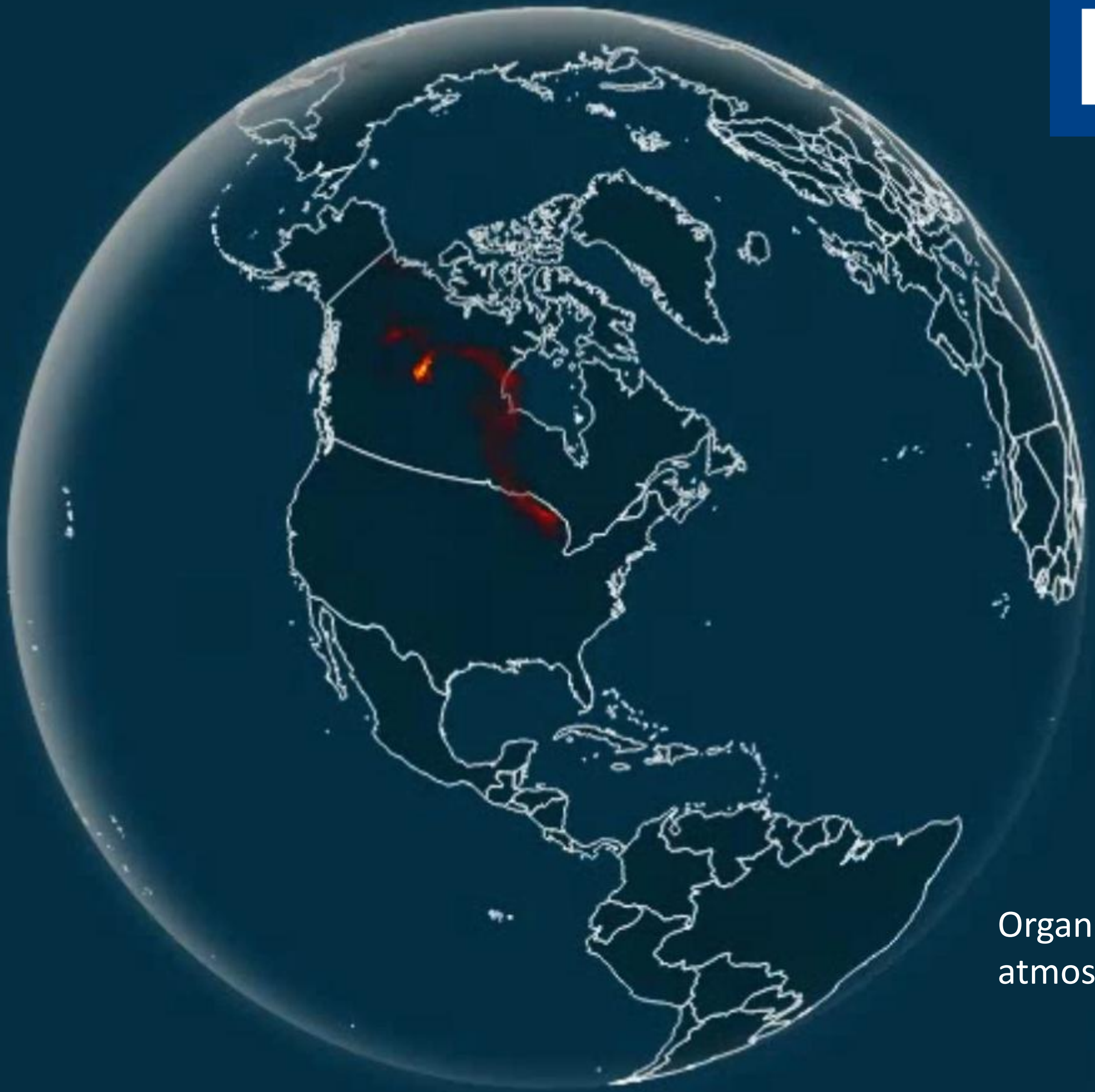
As wildfires are becoming more frequent so is the increase in organic carbon in the atmosphere…



Organic carbon in the atmosphere

2023–08–01 04:
Created by Itohar

https://gmao.gsfc.nasa.gov/

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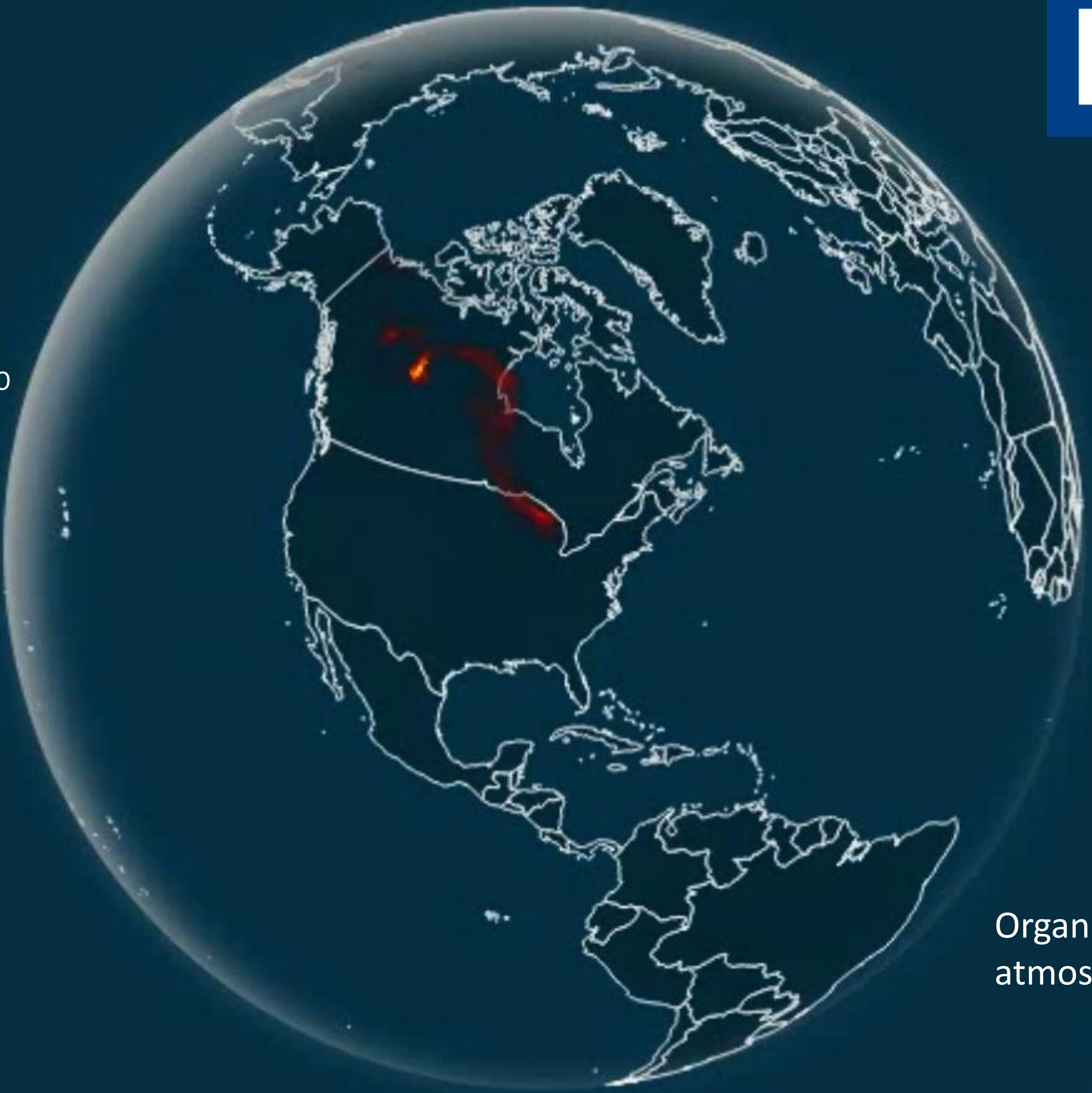
Wildfire smoke, made up largely of dark carbon particles, can block some sunshine from reaching the ground. (Heimerl, 2018.)

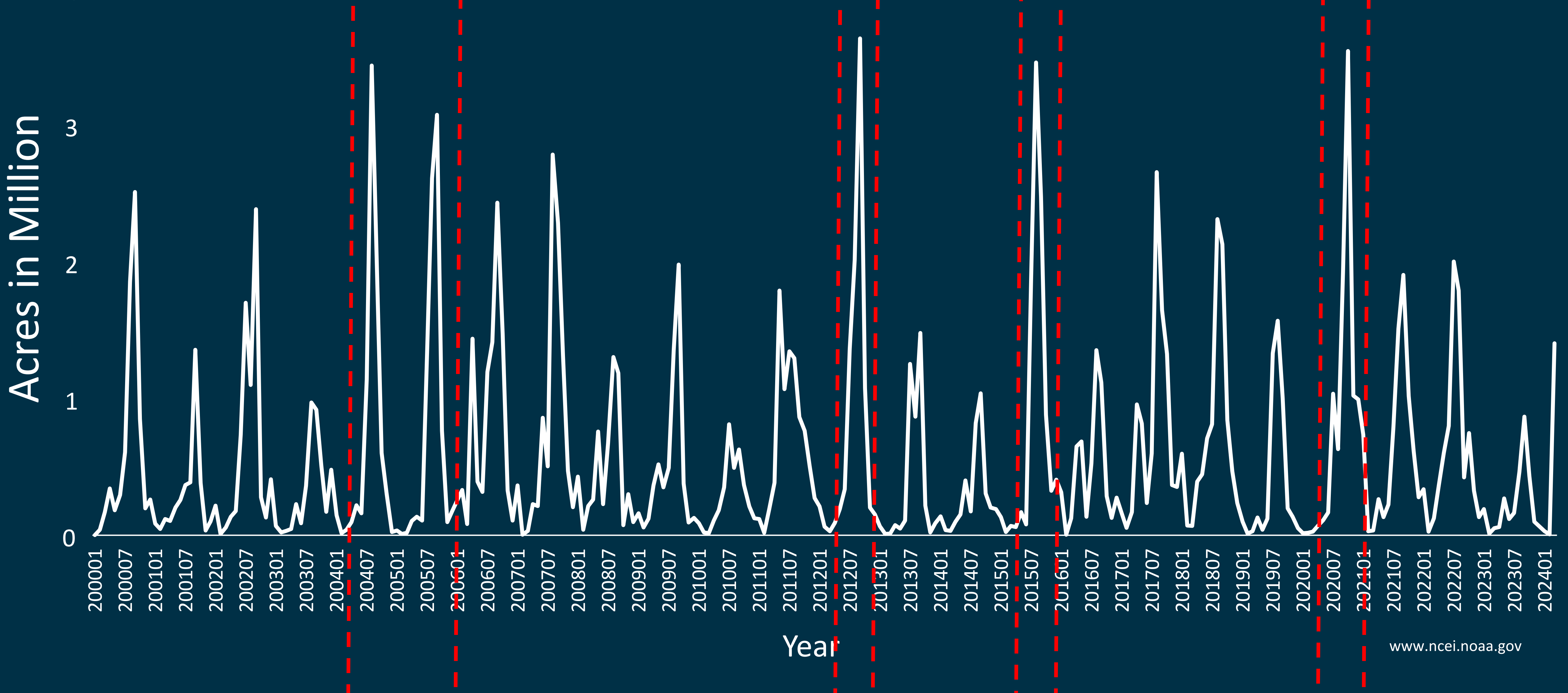


Organic carbon in the atmosphere

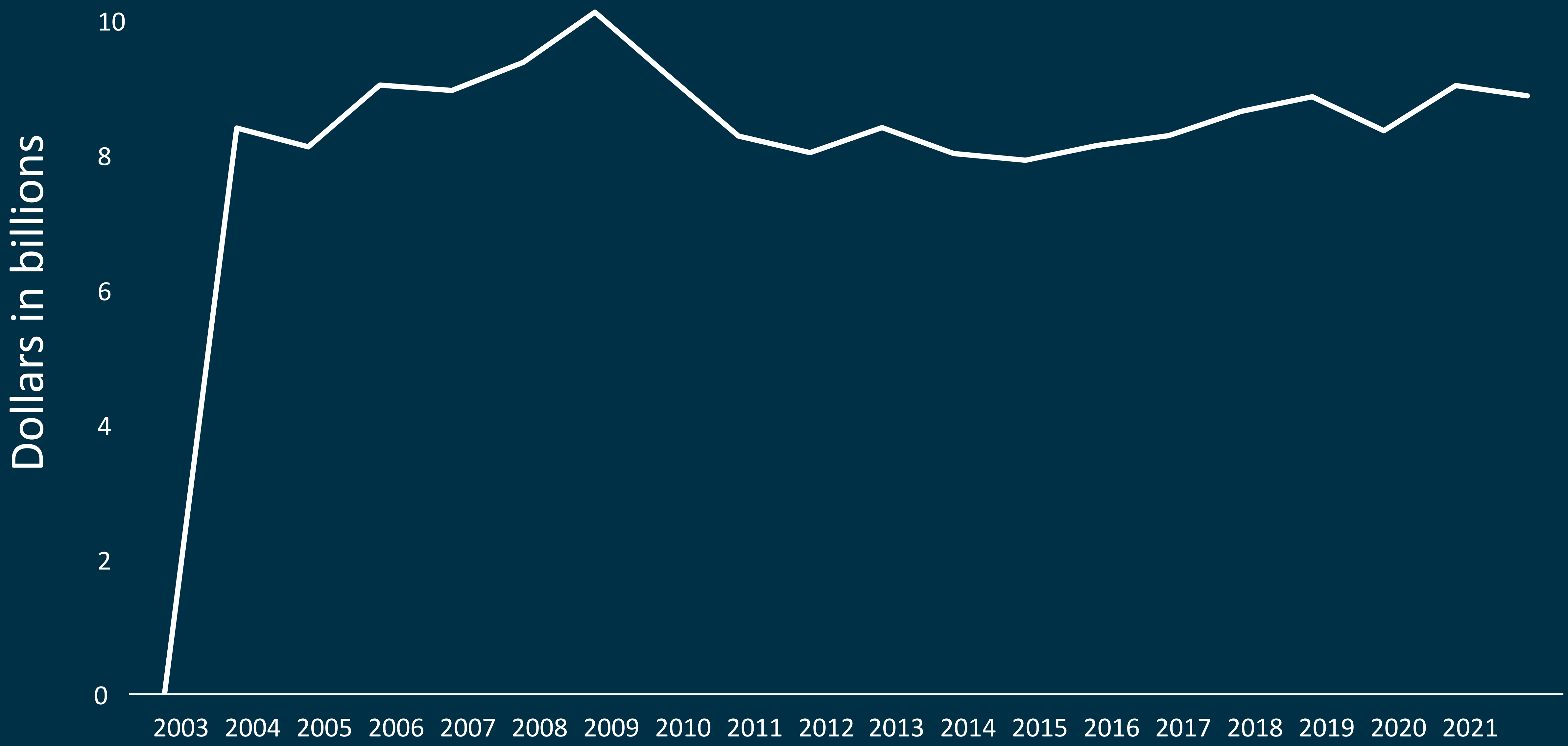
2023-08-01 04:
Created by Itohar
https://gmao.gsfc.nasa.gov/


komonews.com/news/nation-world/

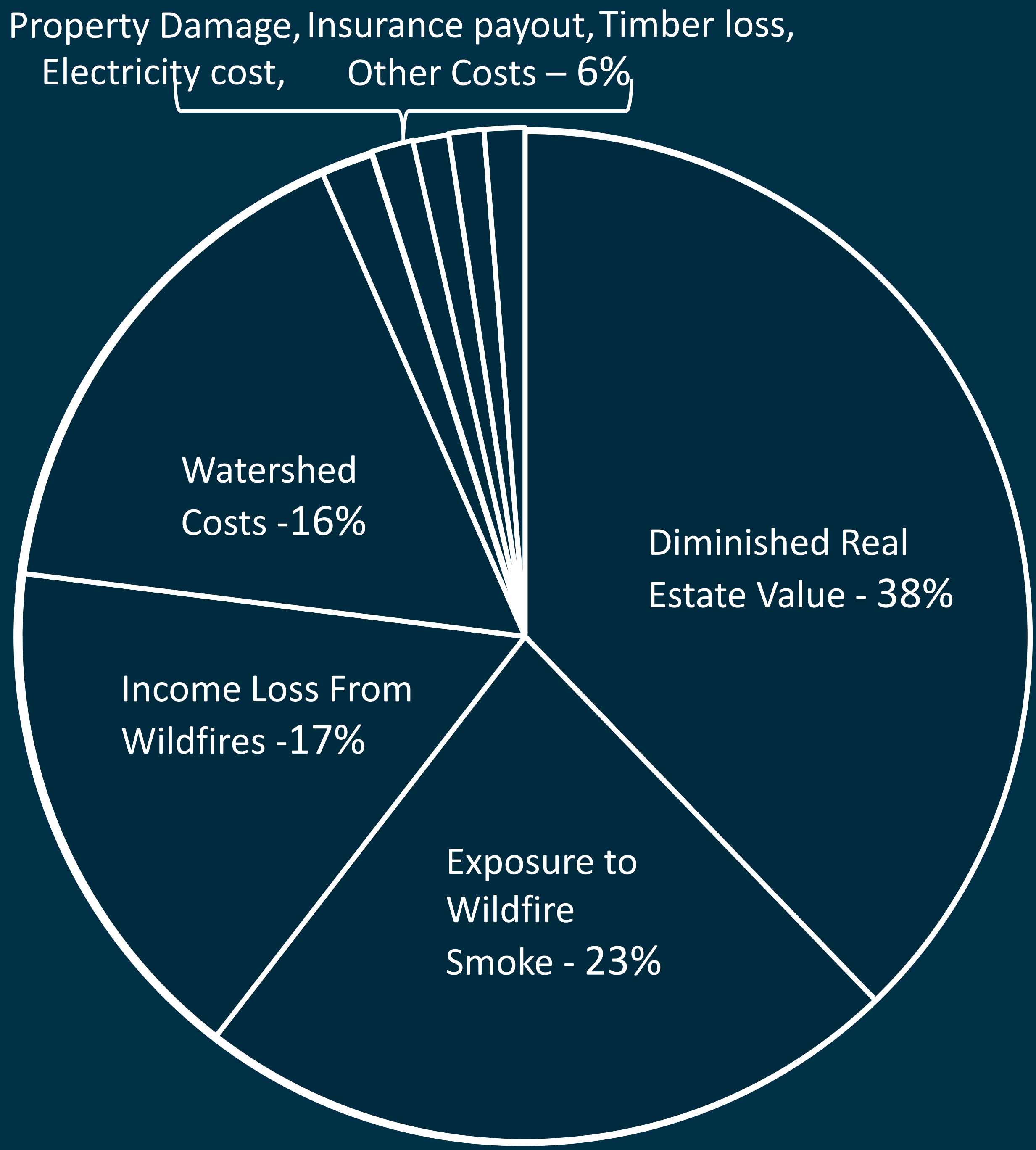
Over time, wildfires have damaged lands in the U.S..



The cost of wildfires in the U.S...

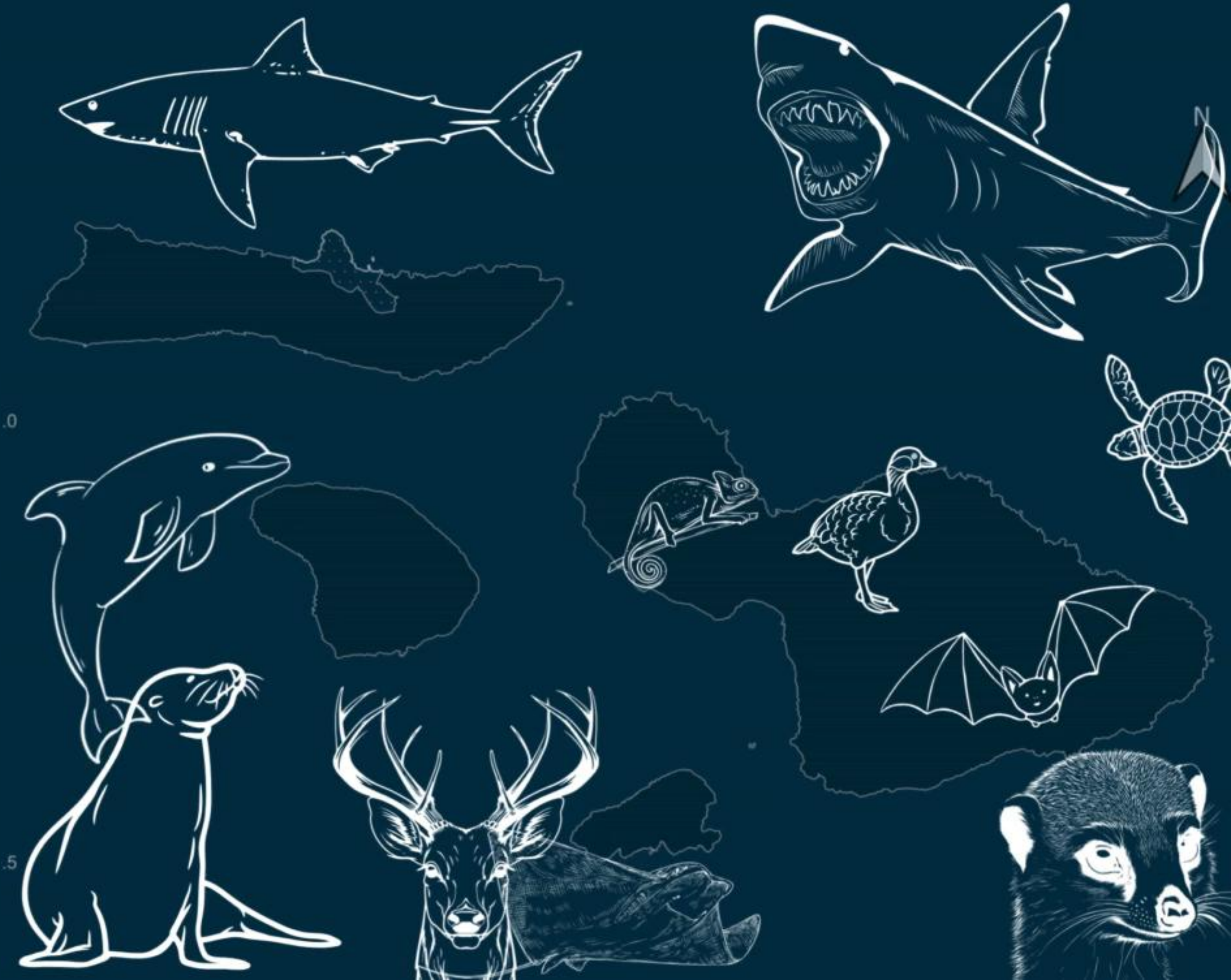


https://www.usfa.fema.gov/statistics/data-sets/



www.jec.senate.gov/

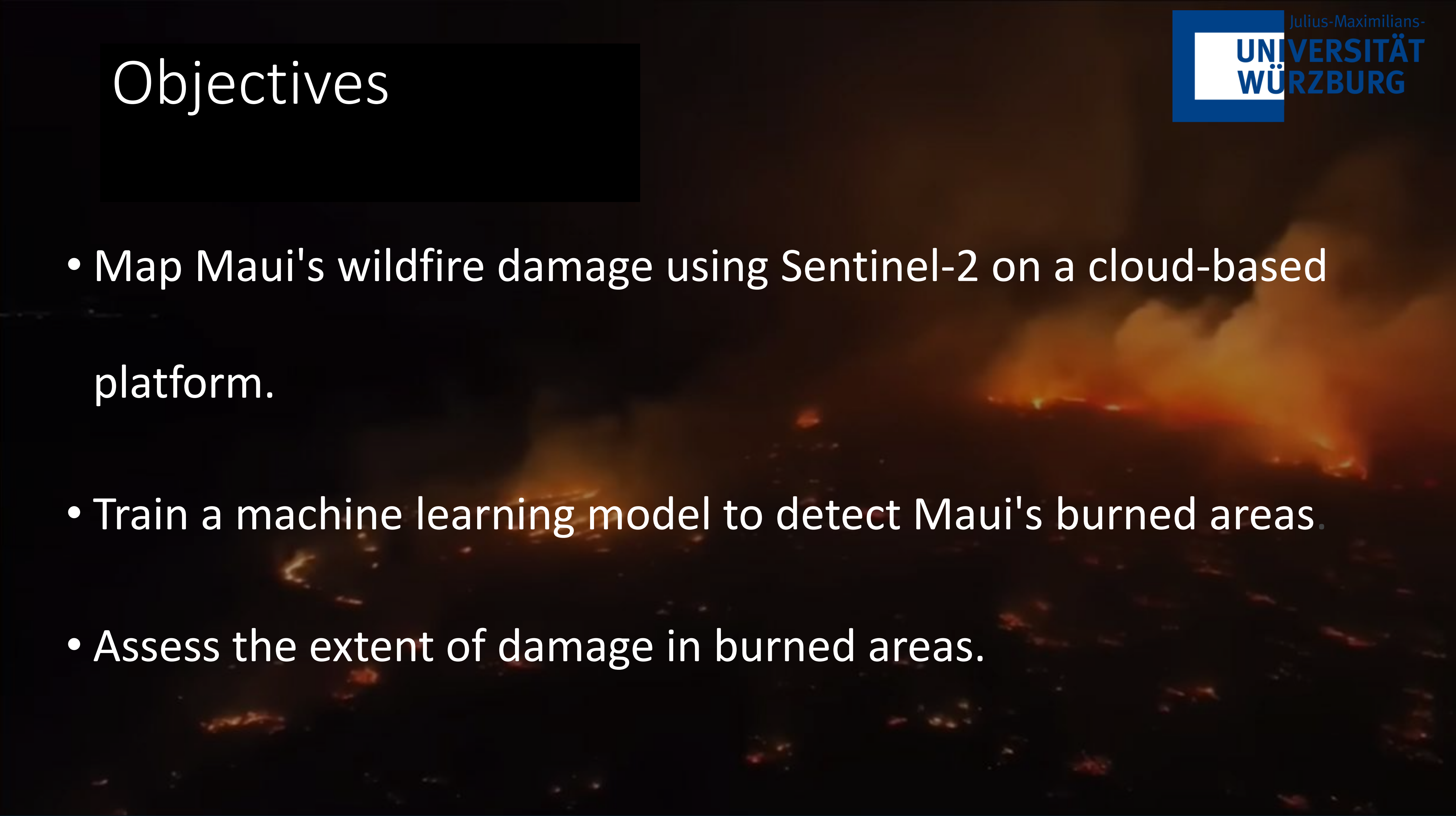
One of the most devastating events in 2023 was the Maui wildfires.



Jesse G. Wald

What contributed to the severity of the fires?

NOAASatellites/status/16893081833



Objectives

- Map Maui's wildfire damage using Sentinel-2 on a cloud-based platform.
- Train a machine learning model to detect Maui's burned areas.
- Assess the extent of damage in burned areas.

Study Area

The Maui county

Datasets used in the Analysis…

Sentinel-2

- Revisits the same location every 5 days.

- Has a resolution as fine as 10 meters.

WorldPOP & ESA-Landcover

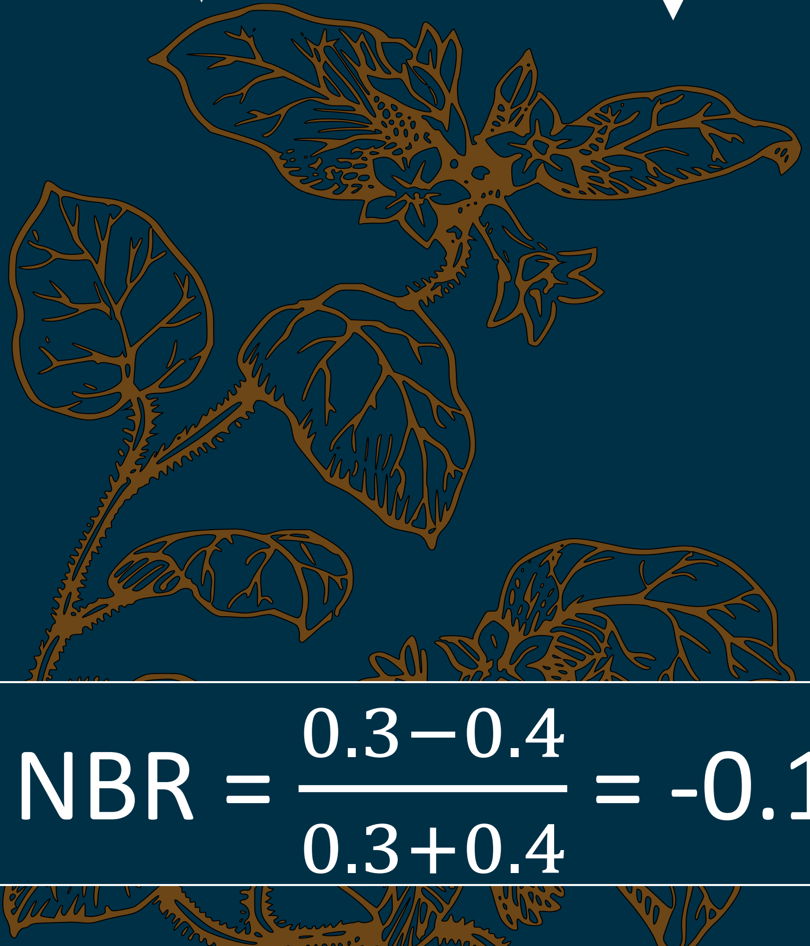
- WorldPOP provides high-resolution population distribution maps at a spatial resolution of 100 meters
- ESA-Landcover offers global land cover maps available at a 10-meter resolution

Normalized Burn Ratio (NBR)

Burned Vegetation

NIR 30%

SWIR 40%



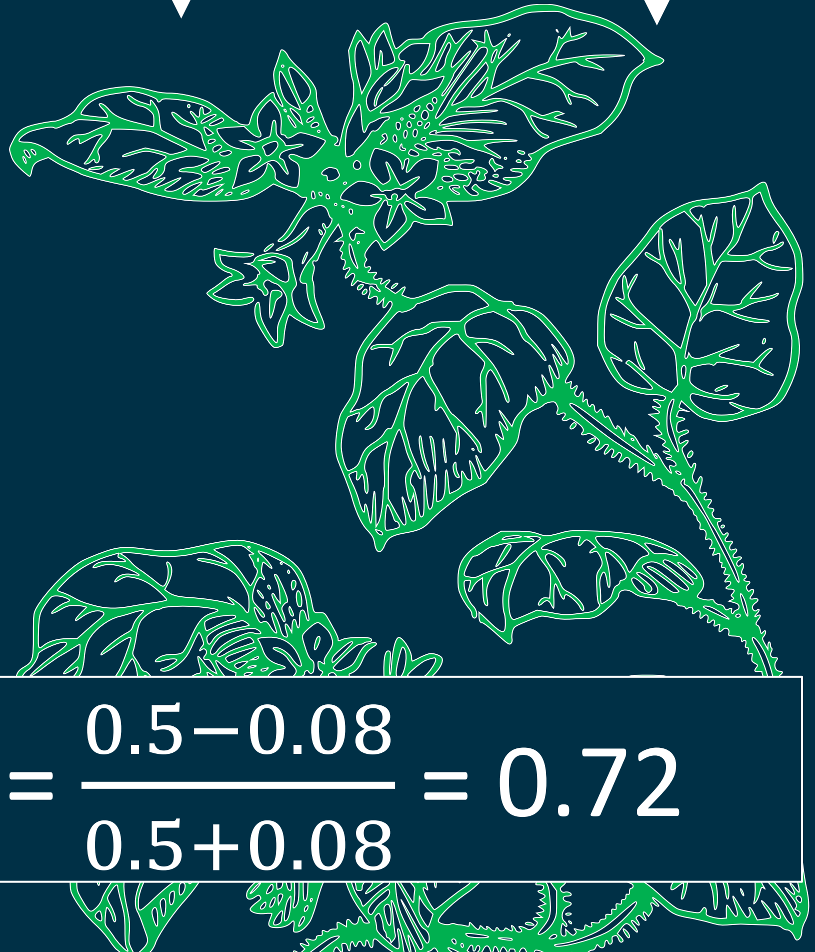
$$NBR = \frac{0.3-0.4}{0.3+0.4} = -0.14$$

$$NBR = \frac{NIR-SWIR}{NIR+SWIR}$$

Healthy Vegetation

NIR 50%

SWIR 8%



$$NBR = \frac{0.5-0.08}{0.5+0.08} = 0.72$$

Normalized Difference Vegetation Index (NDVI)

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Unhealthy Vegetation

NIR 40%

RED 90%

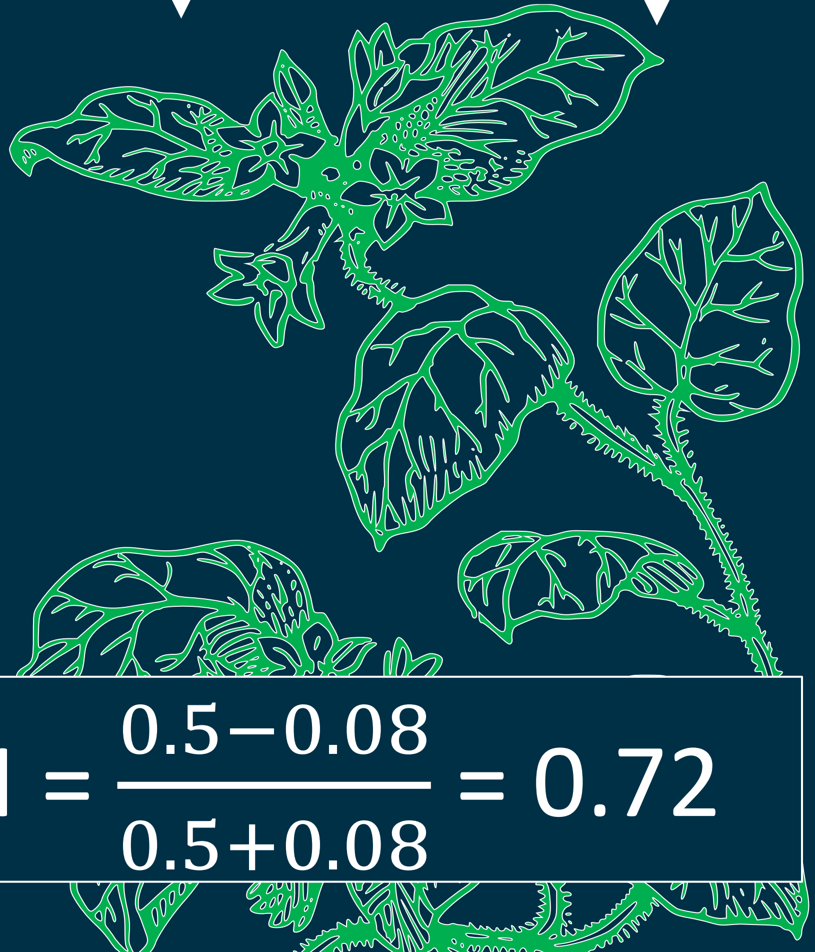
$$NDVI = \frac{NIR-RED}{NIR+RED}$$

$$NDVI = \frac{0.4-0.9}{0.4+0.9} = -0.38$$

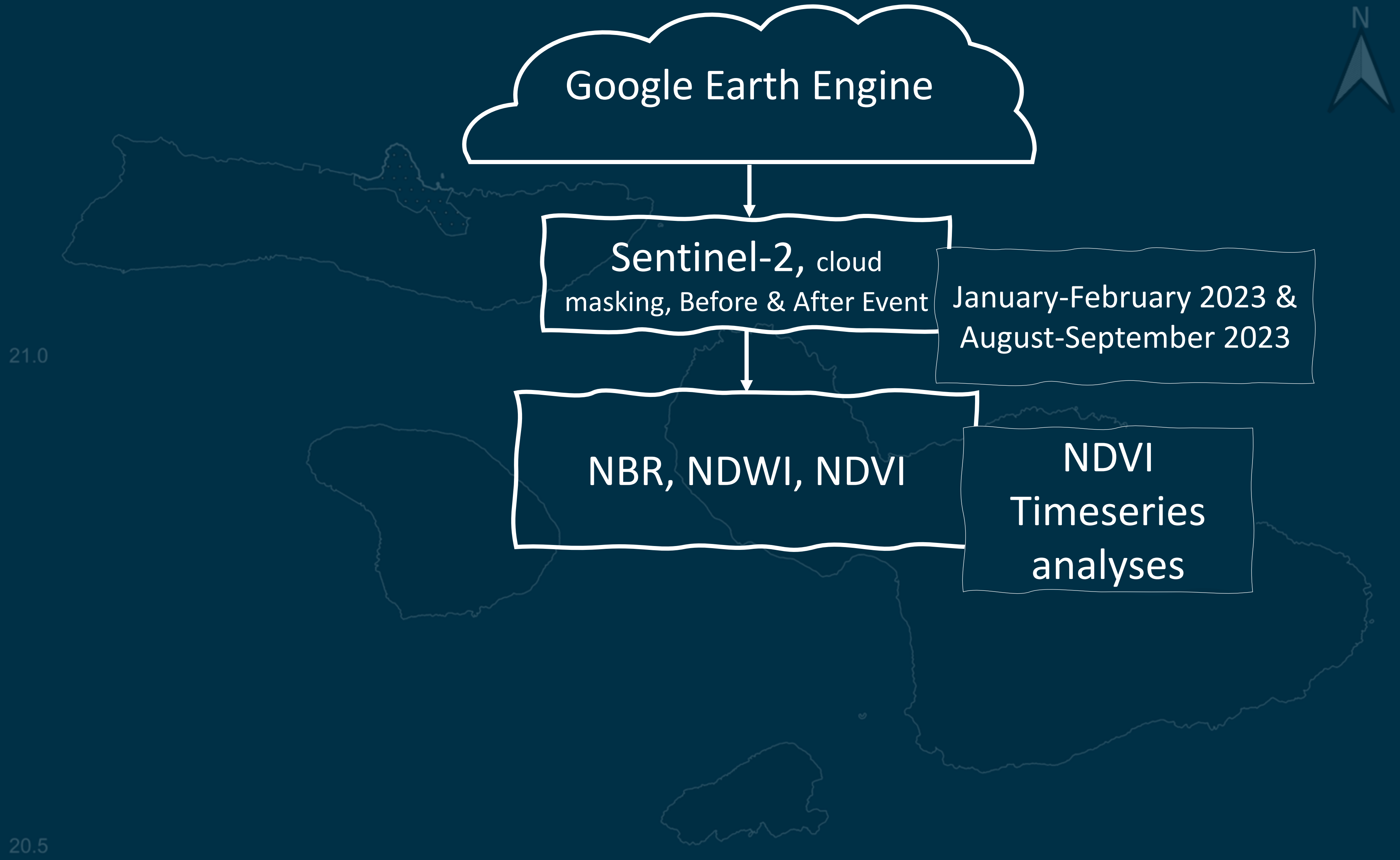
Healthy Vegetation

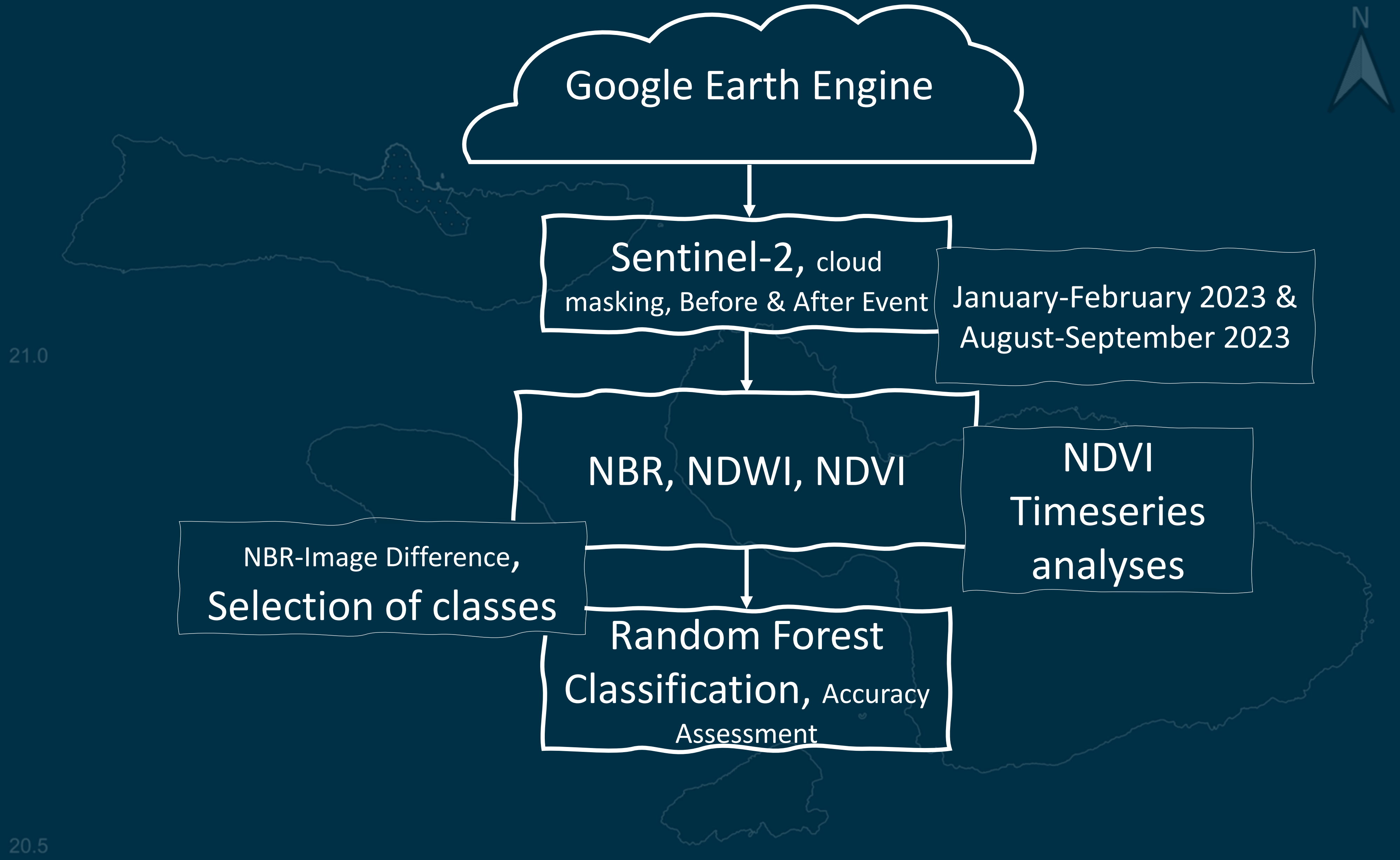
NIR 50%

RED 8%

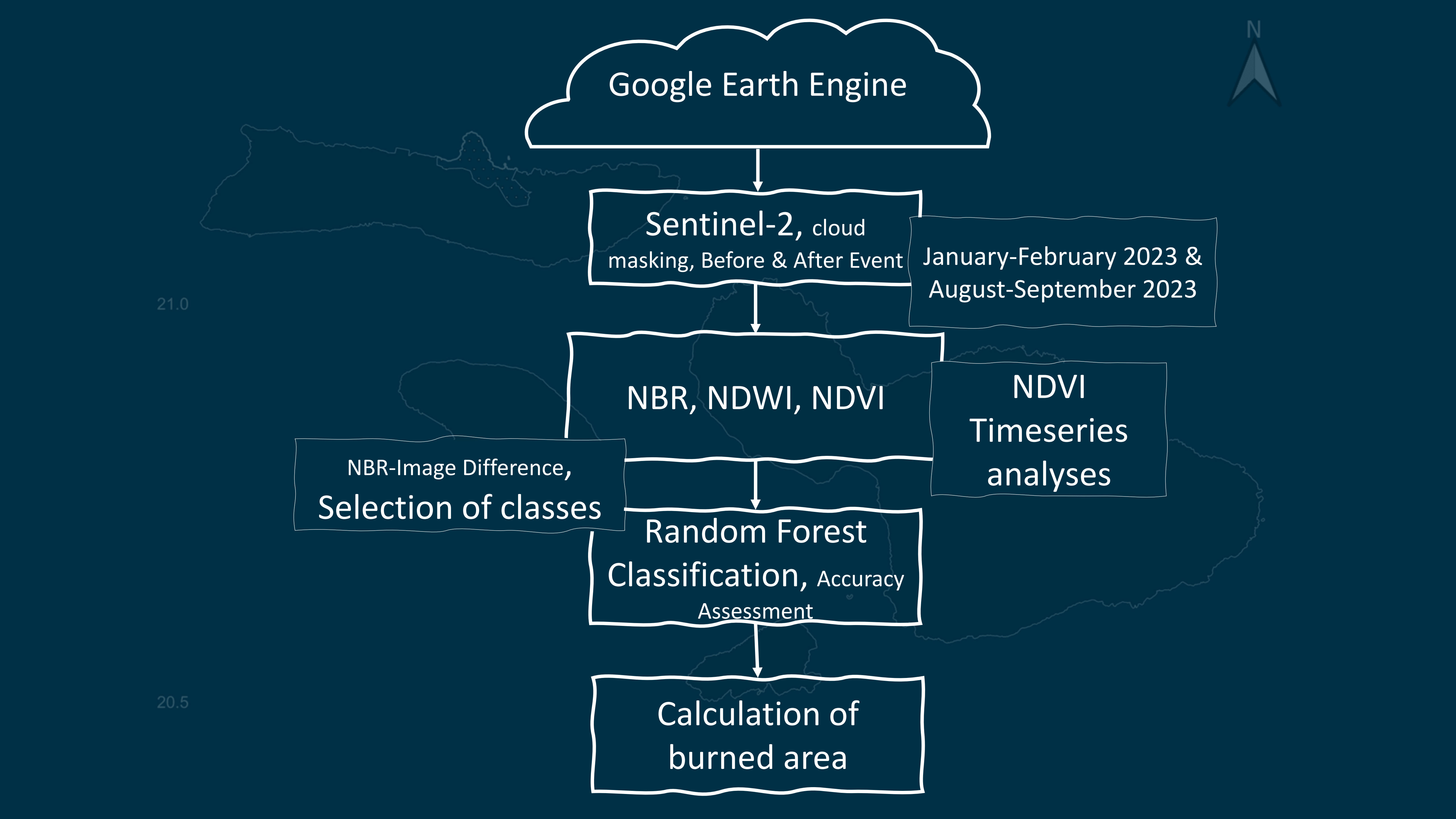


$$NDVI = \frac{0.5-0.08}{0.5+0.08} = 0.72$$


Google Earth Engine
Sentinel-2, cloud masking, Before & After Event
January-February 2023 & August-September 2023
NBR, NDWI, NDVI
NDVI Timeseries analyses
N
21.0
20.5



Google Earth Engine
Sentinel-2, cloud masking, Before & After Event
January-February 2023 & August-September 2023
NBR, NDWI, NDVI
NDVI Timeseries analyses
NBR-Image Difference, Selection of classes
Random Forest Classification, Accuracy Assessment
21.0
20.5
N



Google Earth Engine
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January-February 2023 & August-September 2023
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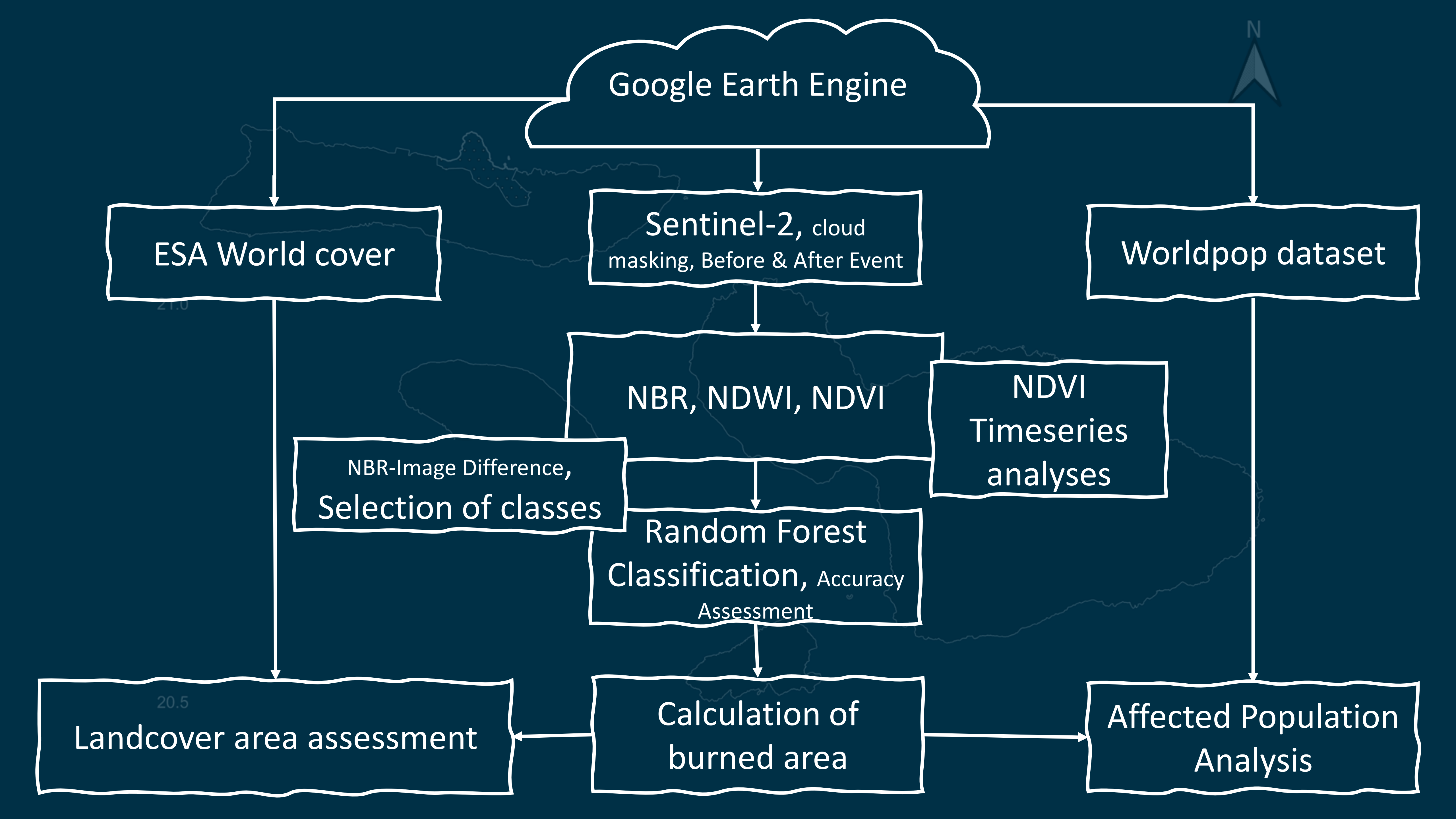
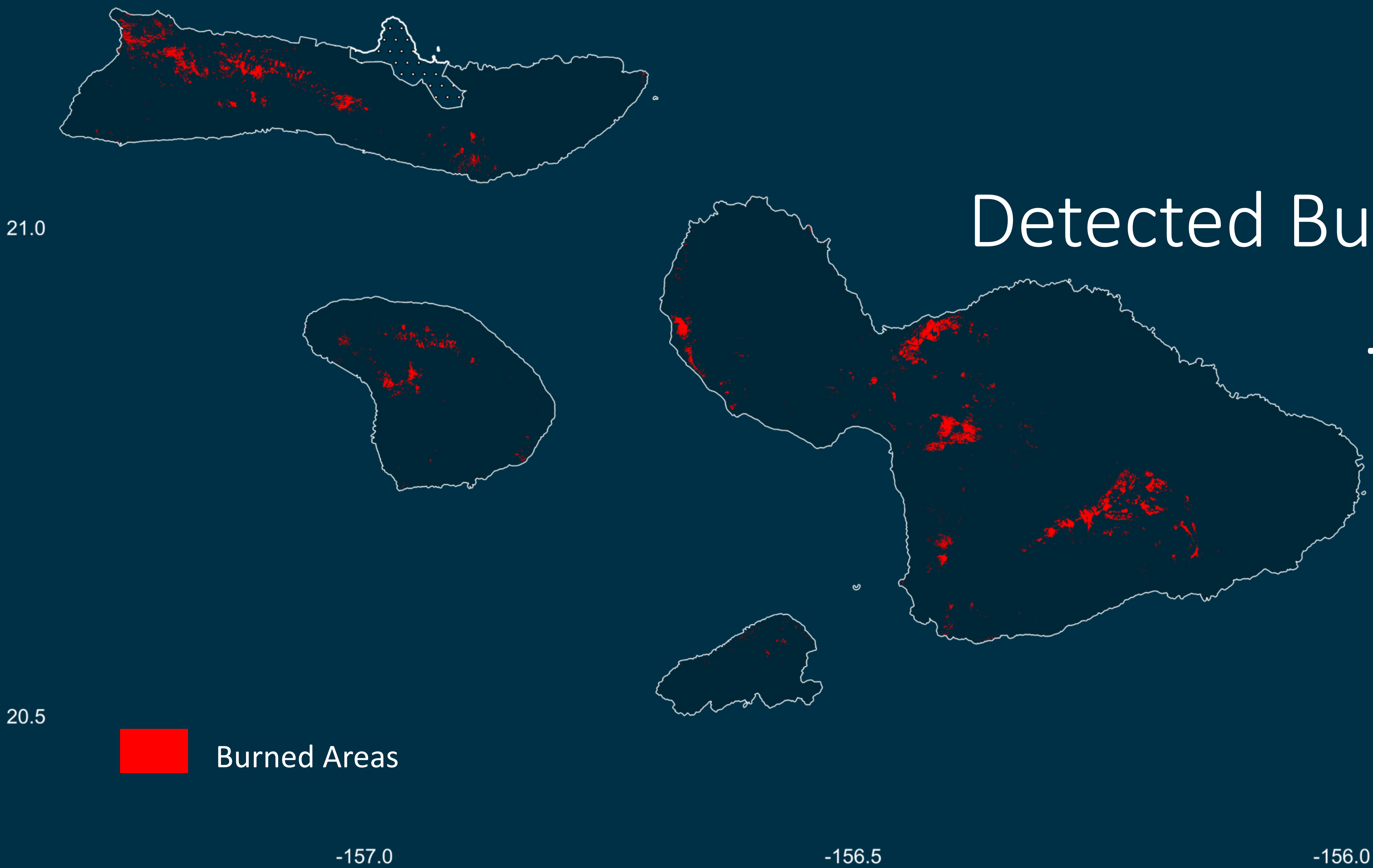
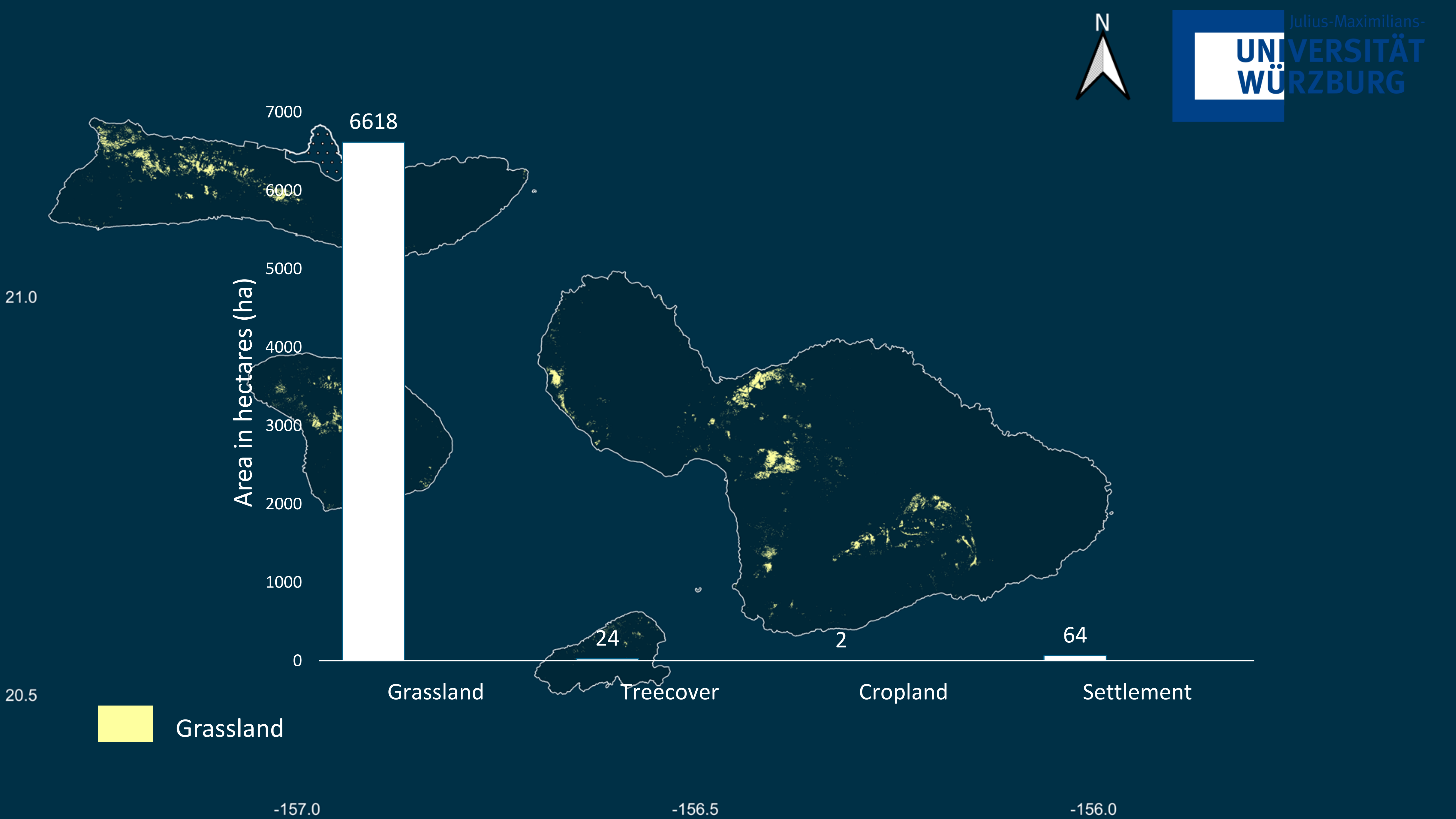
NDVI Timeseries analyses
NBR-Image Difference, Selection of classes
Random Forest Classification, Accuracy Assessment
Calculation of burned area
21.0
20.5
N


Google Earth Engine
ESA World cover
Sentinel-2, cloud masking, Before & After Event
Worldpop dataset
NBR, NDWI, NDVI
NDVI Timeseries analyses
NBR-Image Difference, Selection of classes
Random Forest Classification, Accuracy Assessment
Landcover area assessment
Calculation of burned area
Affected Population Analysis

Detected Burned Areas

7174 ha





N
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6618
Area in hectares (ha)
0
1000
2000
3000
4000
5000
6000
7000
24
2
64
Grassland
Treecover
Cropland
Settlement
21.0
20.5
Grassland
-157.0
-156.5
-156.0

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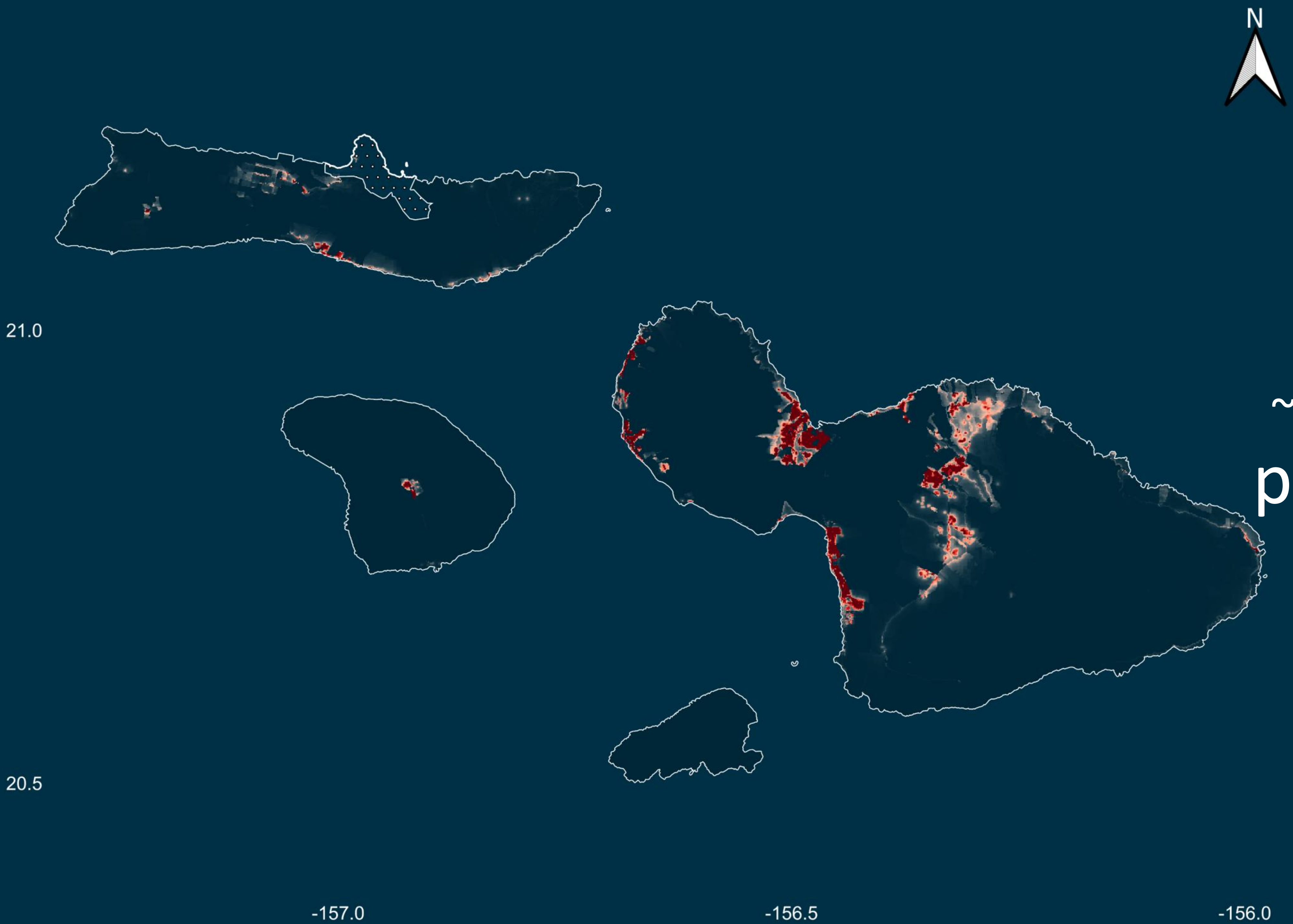
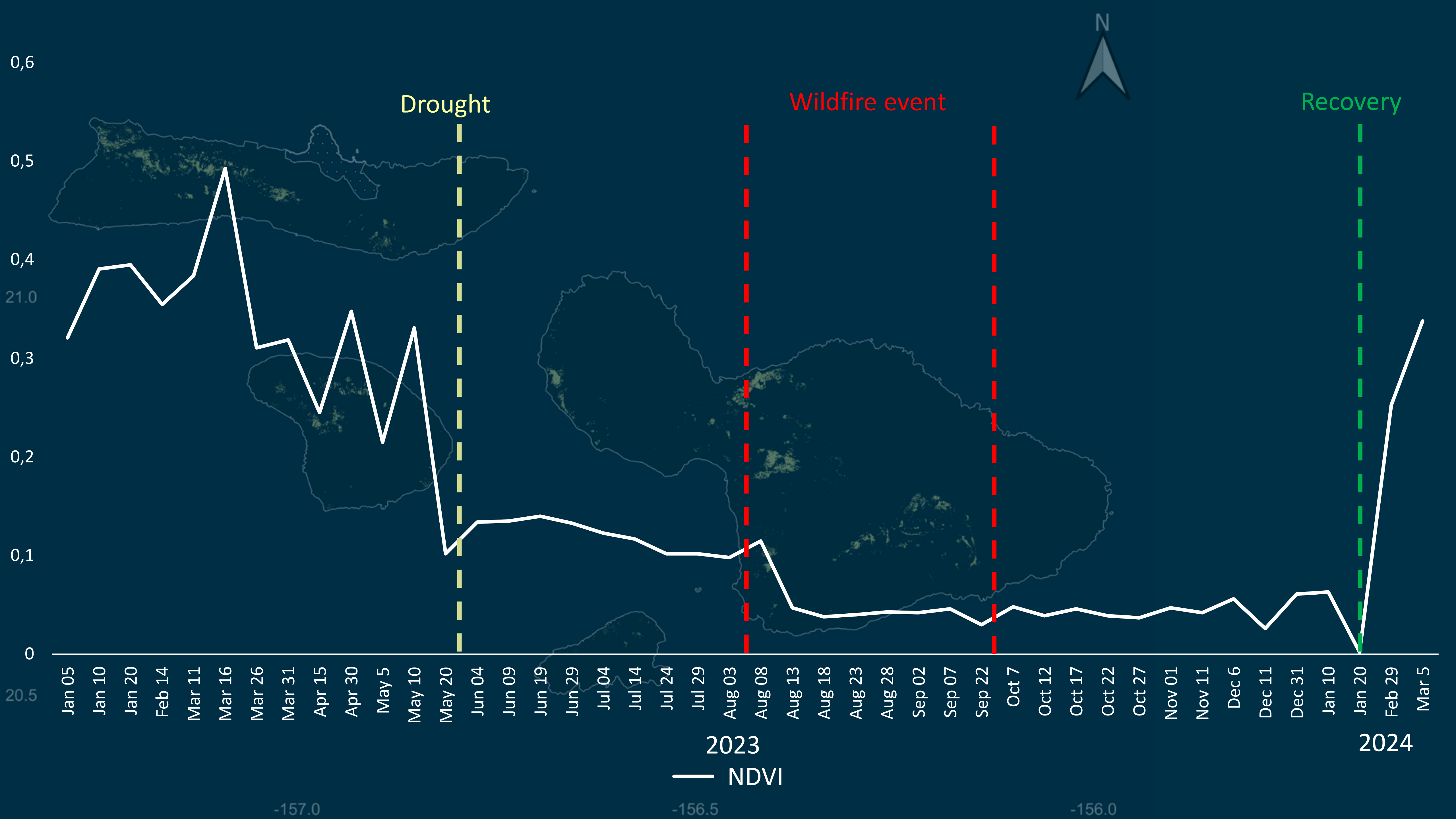


~ 160,000 people were affected

Vegetation cover before, during and after the event...



0,6
0,5
0,4
0,3
0,2
0,1
0
21.0
20.5
-157.0
-156.5
-156.0
N
Drought
Wildfire event
Recovery
Jan 05
Jan 10
Jan 20
Feb 14
Mar 11
Mar 16
Mar 26
Mar 31
Apr 15
Apr 30
May 5
May 10
May 20
Jun 04
Jun 09
Jun 19
Jun 29
Jul 04
Jul 14
Jul 24
Jul 29
Aug 03
Aug 08
Aug 13
Aug 18
Aug 23
Aug 28
Sep 02
Sep 07
Sep 22
Oct 7
Oct 12
Oct 17
Oct 22
Oct 27
Nov 01
Nov 11
Dec 6
Dec 11
Dec 31
Jan 10
Jan 20
Feb 29
Mar 5
2023
2024
NDVI

0.6

NDVI

-157.0 -156.5 -156.0

0.6

N

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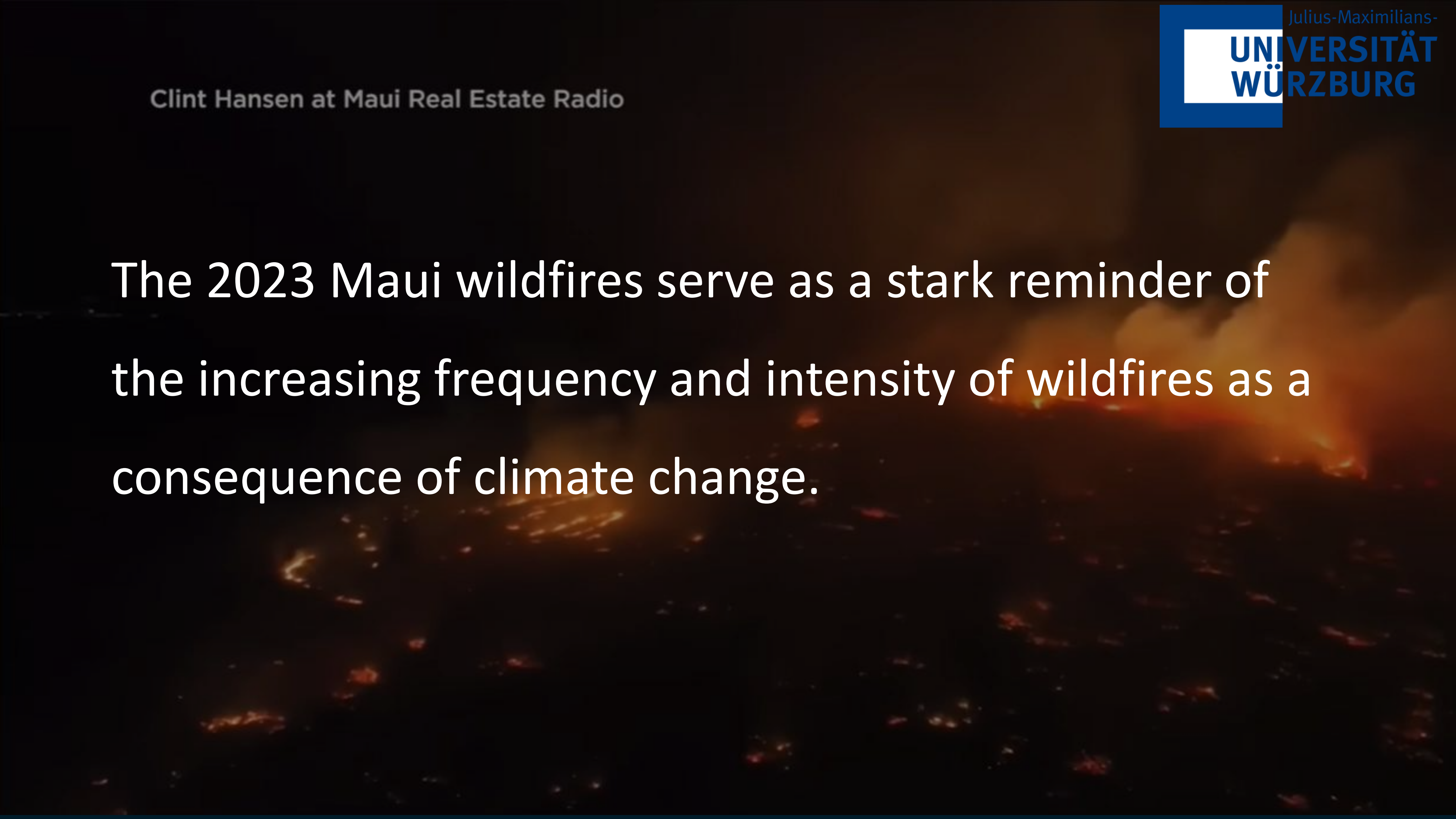
NDVI

-157.0 -156.5 -156.0

In conclusion...

The study underscores the effectiveness of combining remote sensing and machine learning in disaster assessment and management.


Clint Hansen at Maui Real Estate Radio
Julius-Maximilians-
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WÜRZBURG
The 2023 Maui wildfires serve as a stark reminder of the increasing frequency and intensity of wildfires as a consequence of climate change.

Thank you for your attention

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